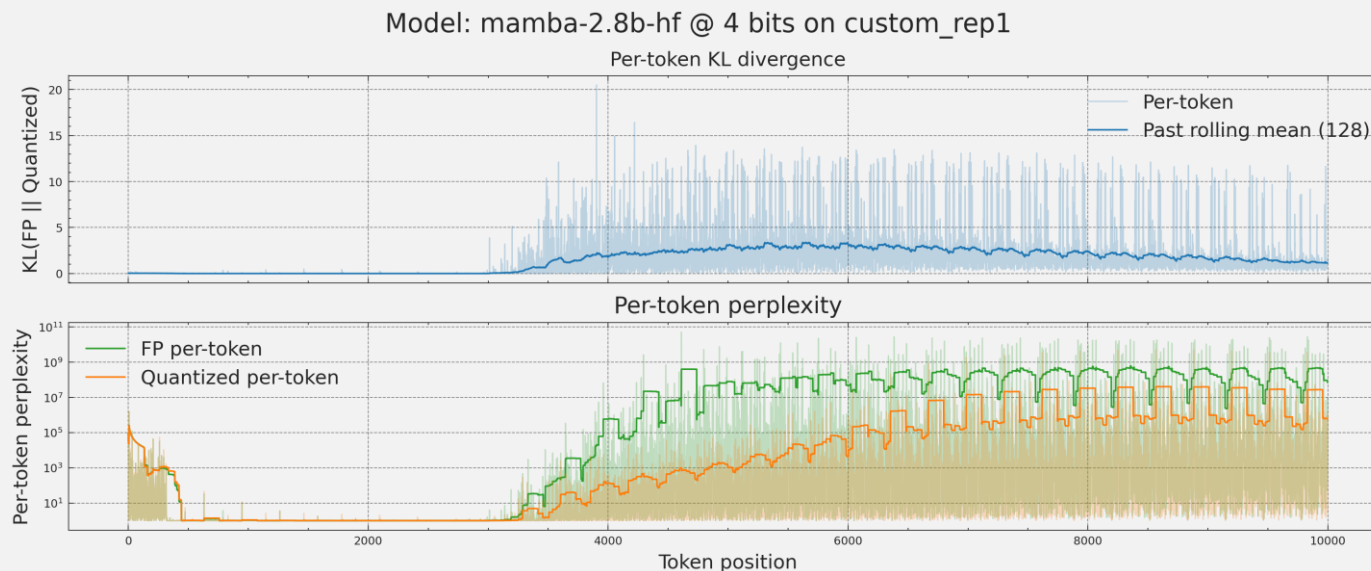

SSM Optimization

Weekly Meeting

15/05/26

Longer sequences

- After the results of last week, I evaluated on longer sequences



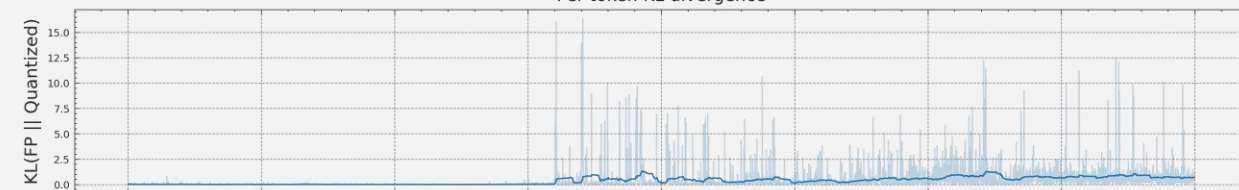
- Considering Mamba-2.8b, the gap seems to converge to a stable value after token ~8000
- Trend between tokens 400 and 3000 seems reasonable

Longer sequences

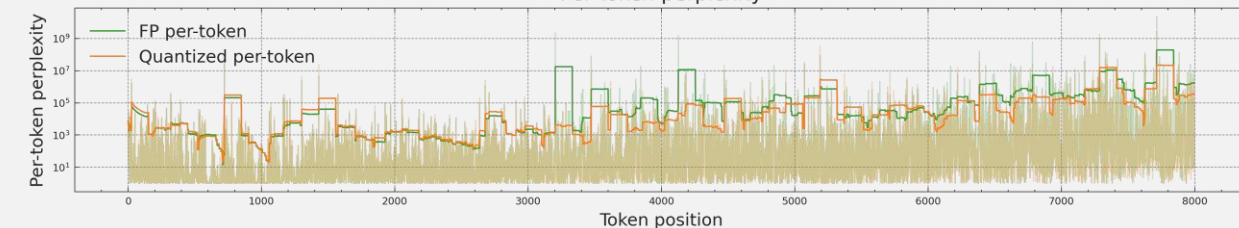
- On Wikitext2 and C4, with a sequence length of 8000 tokens:

Model: mamba-2.8b-hf @ 4 bits on wikitext2_0

Per-token KL divergence

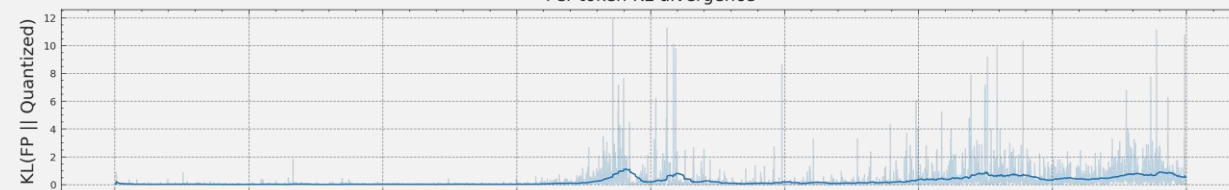


Per-token perplexity

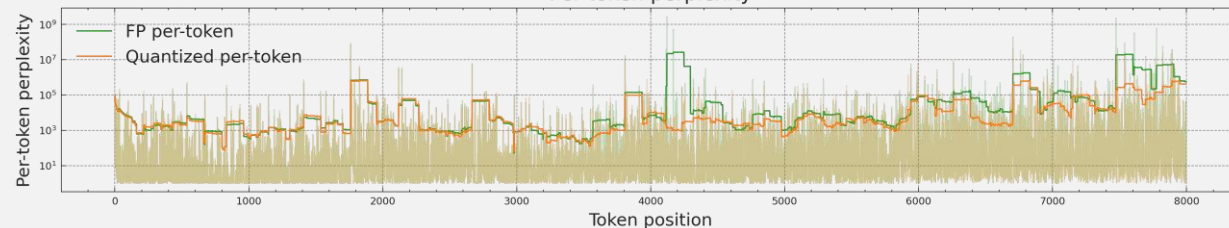


Model: mamba-2.8b-hf @ 4 bits on c4_0

Per-token KL divergence

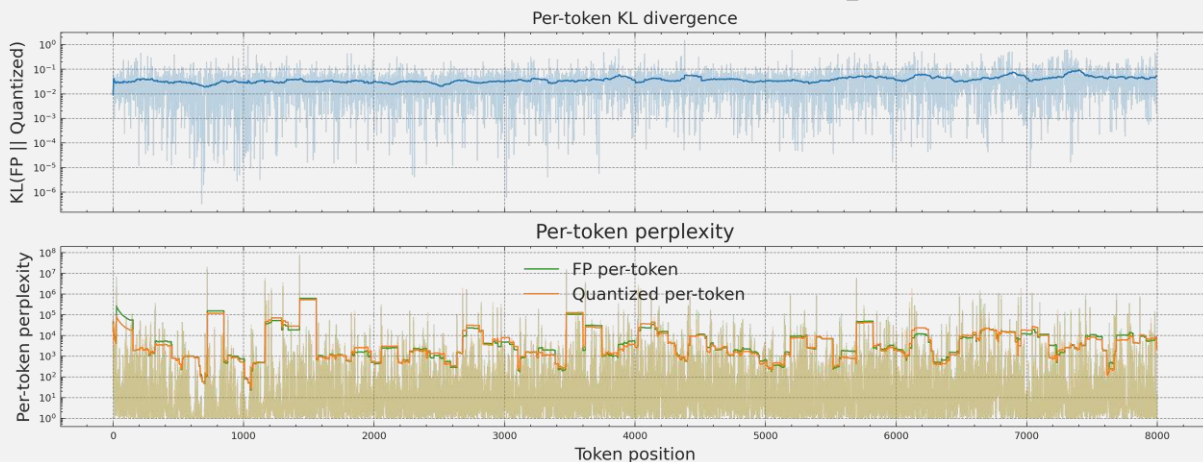


Per-token perplexity

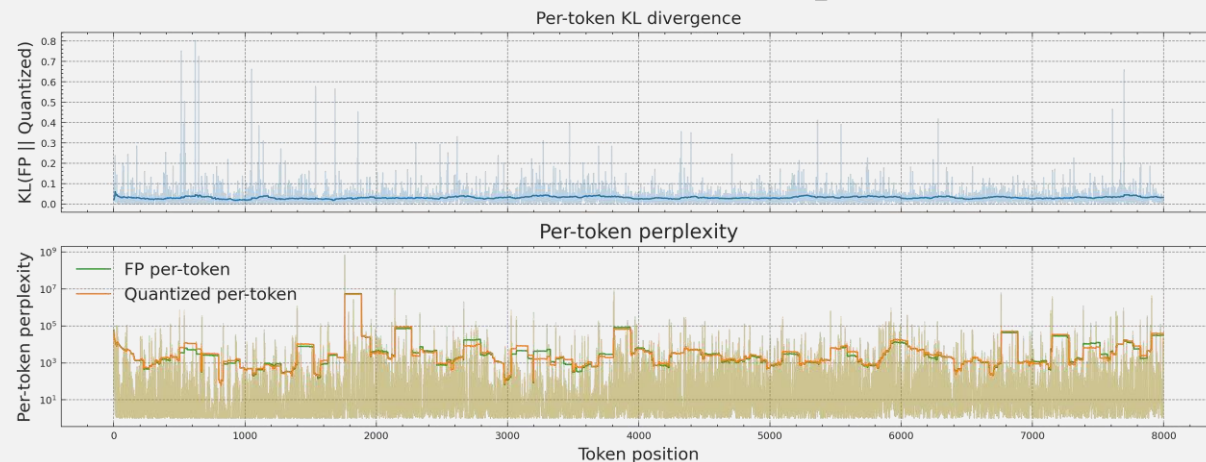


Longer sequences – Mamba-1.4b

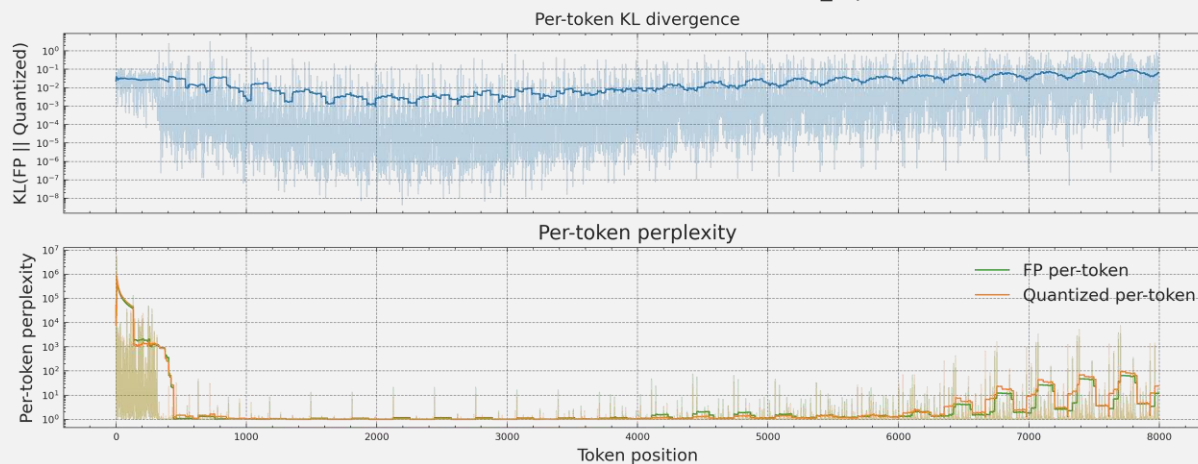
Model: mamba-1.4b-hf @ 4 bits on wikitext2_0



Model: mamba-1.4b-hf @ 4 bits on c4_0

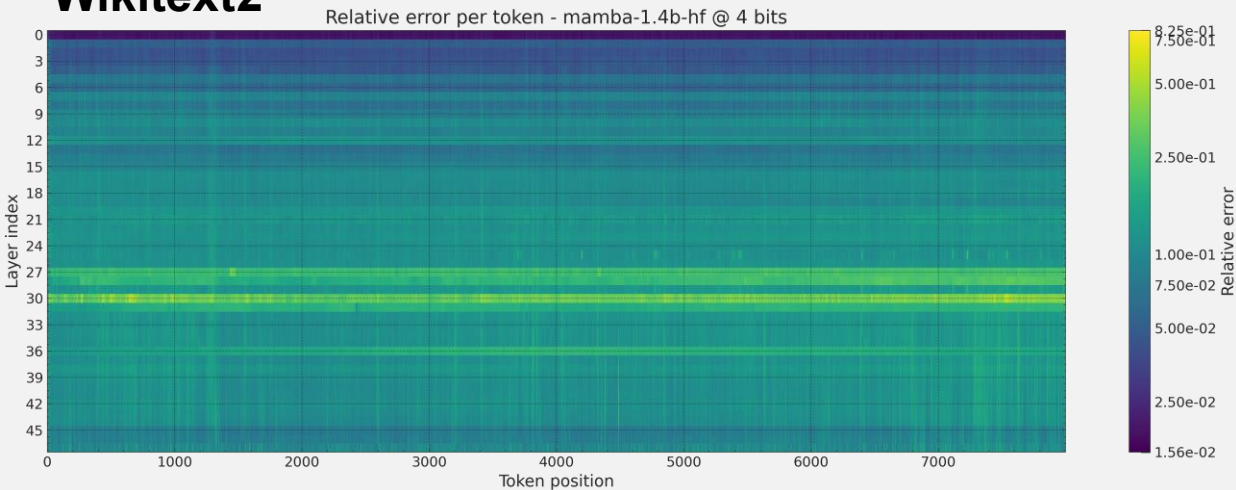


Model: mamba-1.4b-hf @ 4 bits on custom_rep1

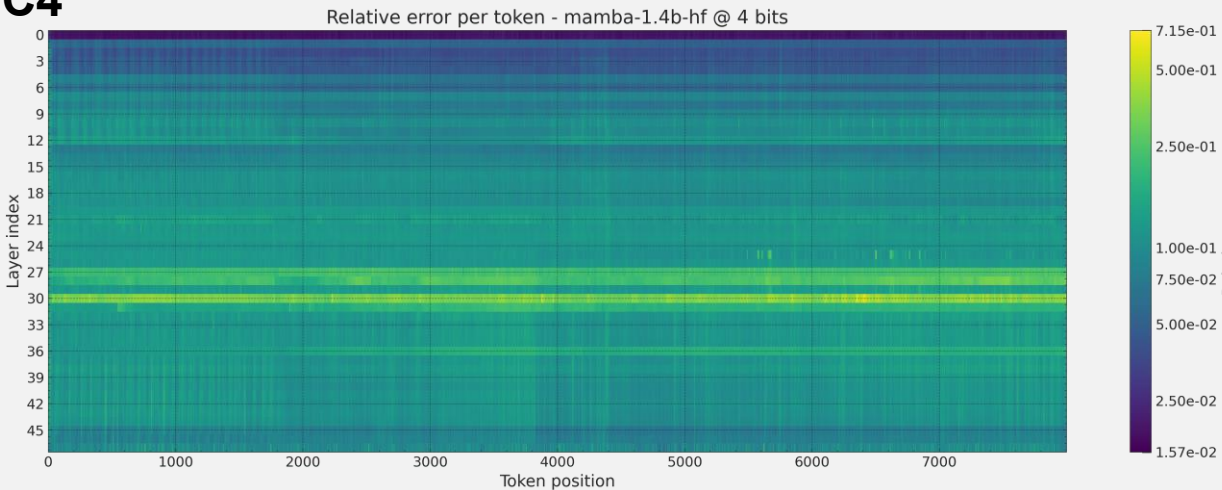


Longer sequences – Mamba-1.4b

Wikitext2



C4

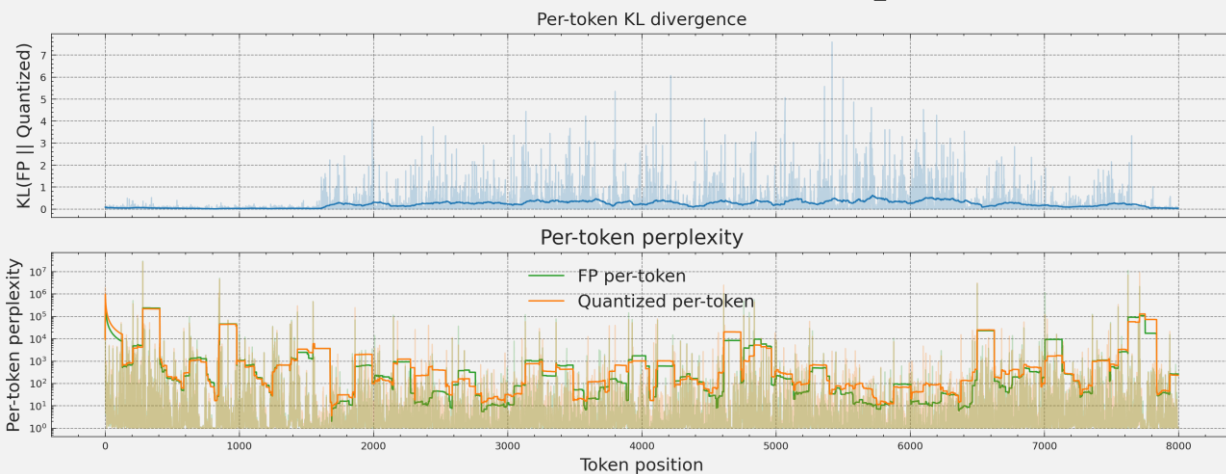


Rep.

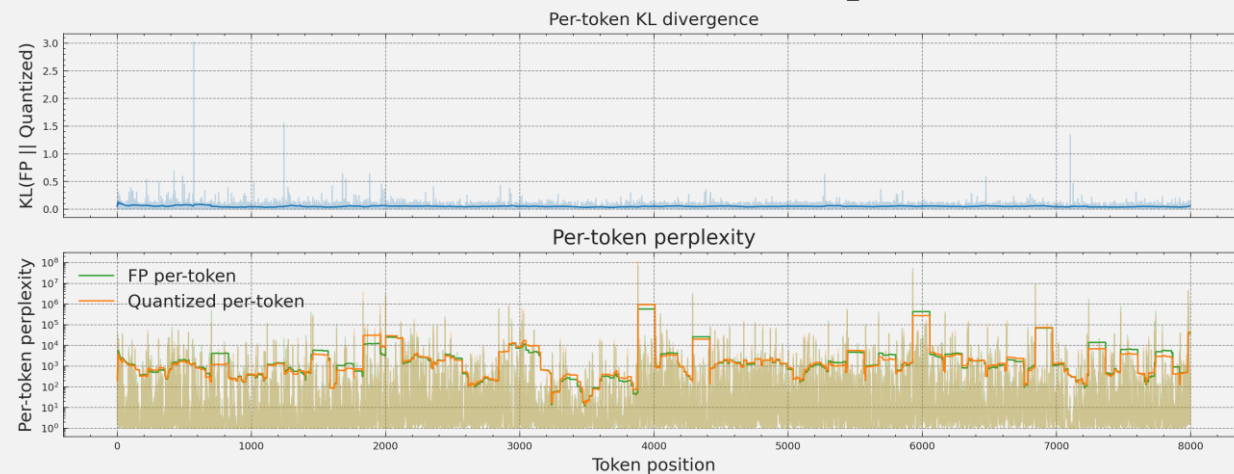


Longer sequences - Falcon-Mamba-7b

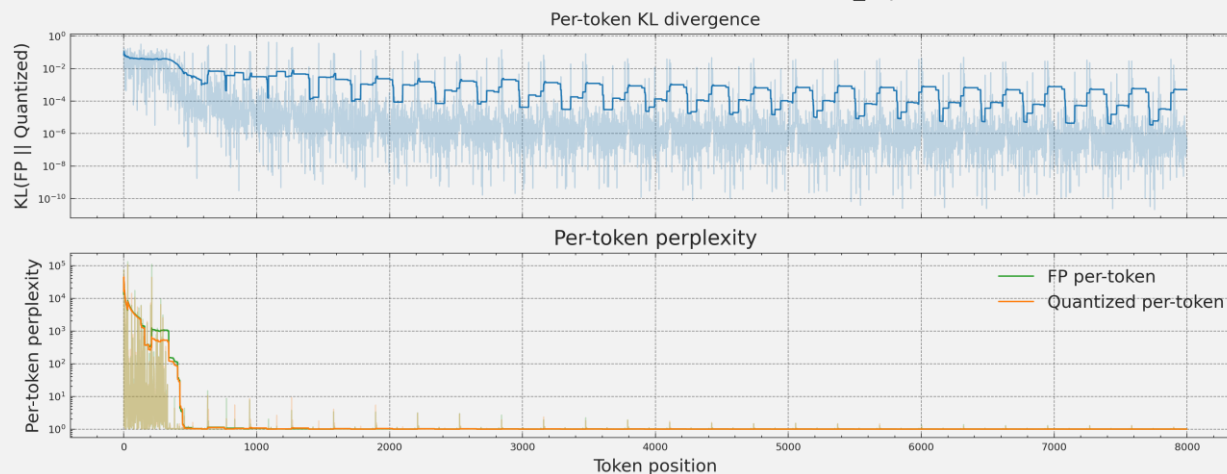
Model: falcon-mamba-7b @ 4 bits on wikitext2_0



Model: falcon-mamba-7b @ 4 bits on c4_0

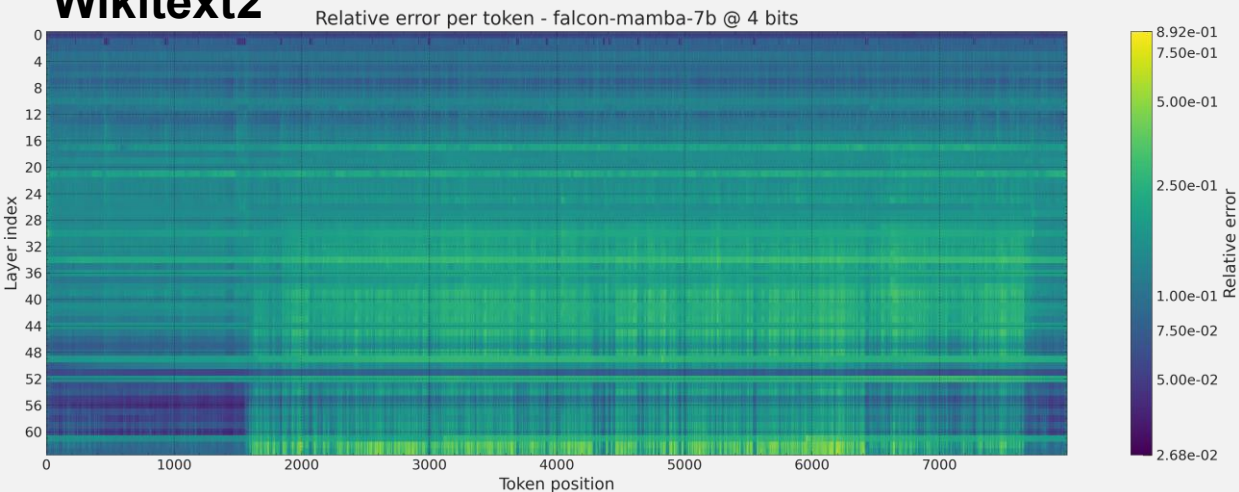


Model: falcon-mamba-7b @ 4 bits on custom_rep1

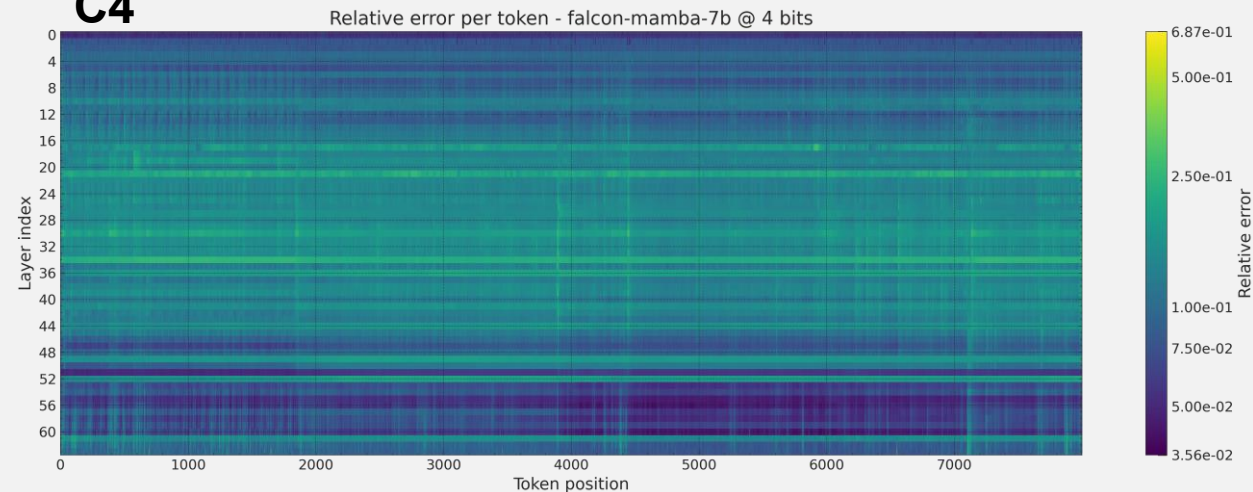


Longer sequences - Falcon-Mamba-7b

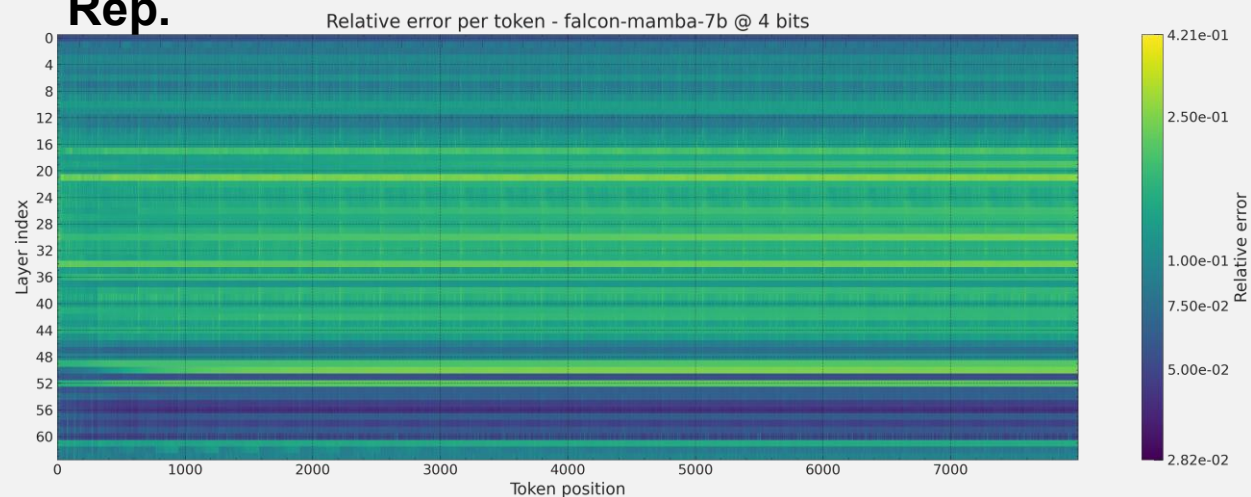
Wikitext2



C4



Rep.



Quantization

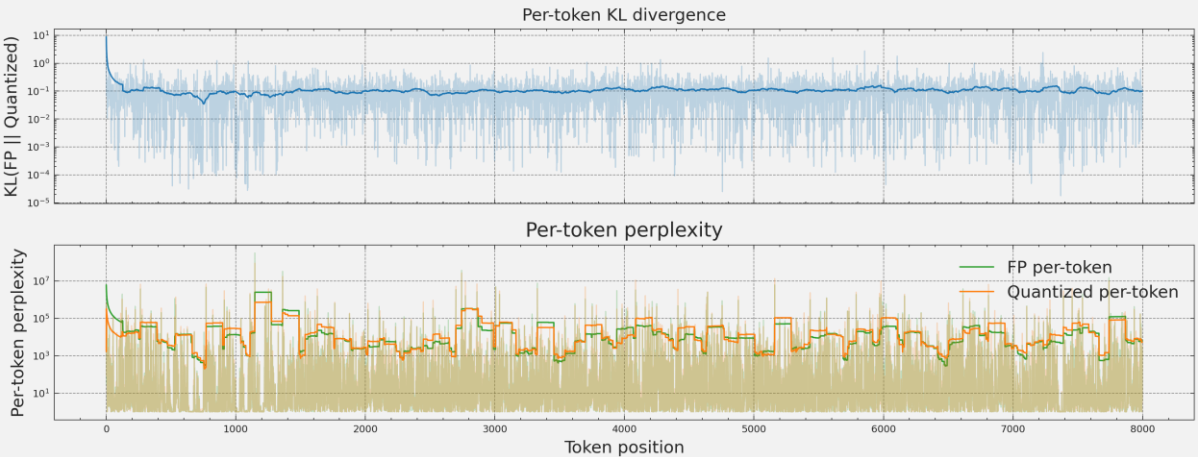
Qwen3.5-0.8B

		Wikitext2							
Model	Quantized	SINQ		RTN		bitsandbytes		HQQ	
		Perplexity	KL divergence	Perplexity	KL divergence	Perplexity	KL divergence	Perplexity	KL divergence
Qwen3.5-0.8B-Base		13.123	-	-	-	-	-	-	-
	 (4 bits)	14.448	0.098	14.689	0.111	16.070	0.205	14.525	0.100
	 (2 bits)	16977.691	7.132	57919.547	8.356	-	-	17484.561	7.145
Qwen3.5-0.8B		17.187	-	-	-	-	-	-	-
	 (4 bits)	18.687	0.108	19.520	0.120	23.775	0.267	19.720	0.114
	 (2 bits)	27562.484	7.476	104045.813	8.893	-	-	19045.381	7.052
C4									
Qwen3.5-0.8B-Base		17.632	-	-	-	-	-	-	-
	 (4 bits)	19.151	0.088	19.336	0.098	20.980	0.182	19.190	0.090
	 (2 bits)	15144.127	6.766	50837.938	7.976	-	-	9211.590	6.266
Qwen3.5-0.8B		20.407	-	-	-	-	-	-	-
	 (4 bits)	22.522	0.097	23.219	0.114	25.375	0.206	22.964	0.104
	 (2 bits)	17638.152	6.986	65605.977	8.356	-	-	9988.983	6.347

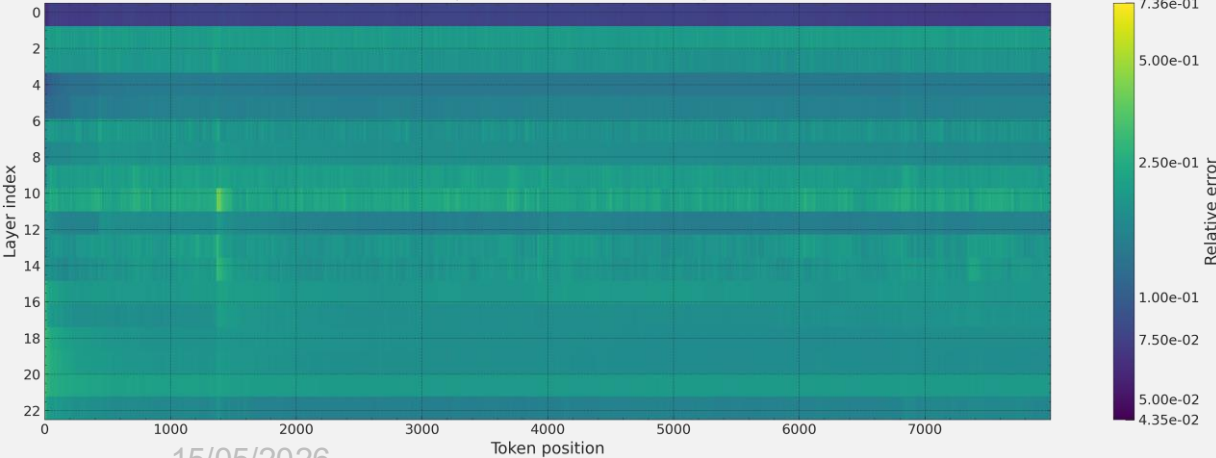
State drift

Longer sequences – Qwen3.5-0.8B

Model: Qwen3.5-0.8B @ 4 bits on wikitext2_0

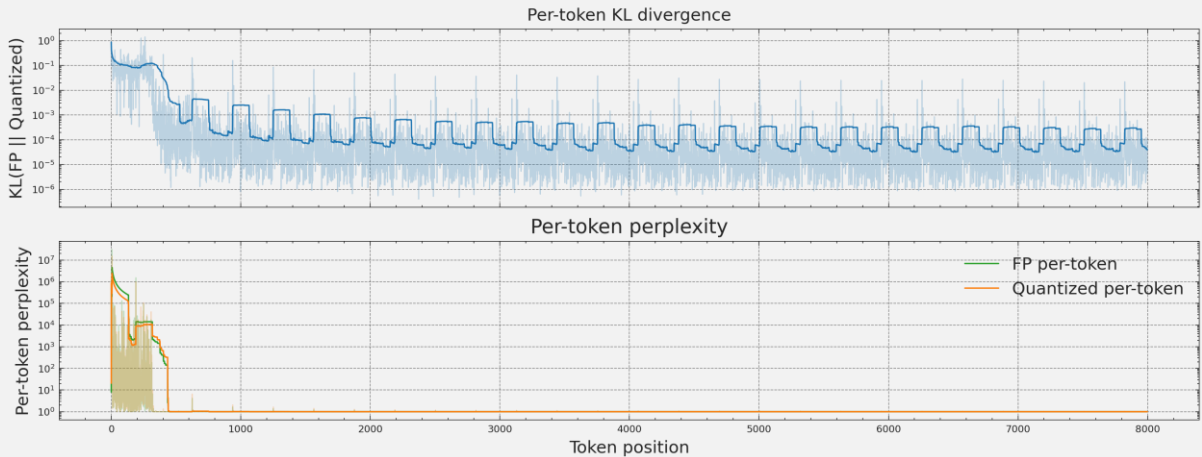


Relative error per token - Qwen3.5-0.8B @ 4 bits

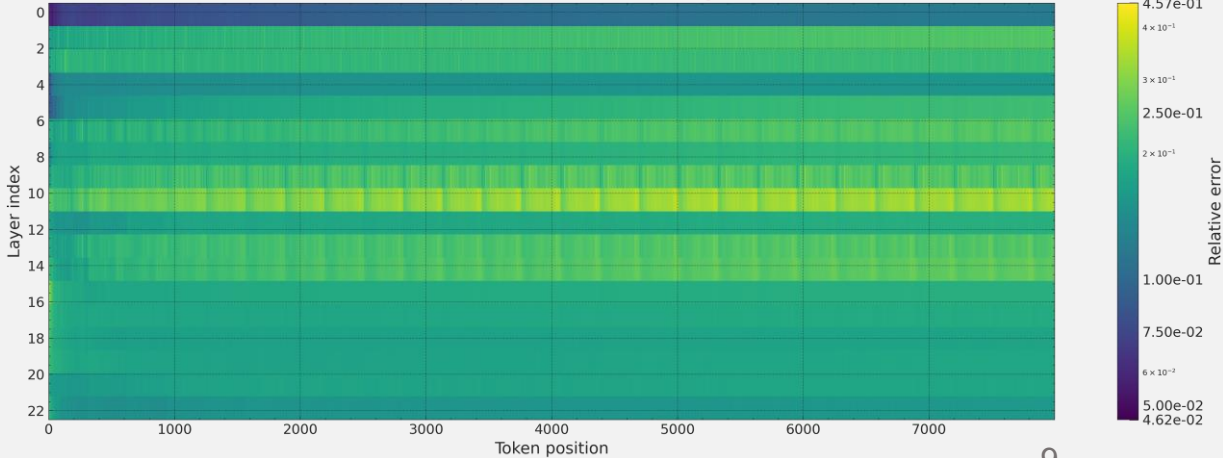


15/05/2026

Model: Qwen3.5-0.8B @ 4 bits on custom_rep1



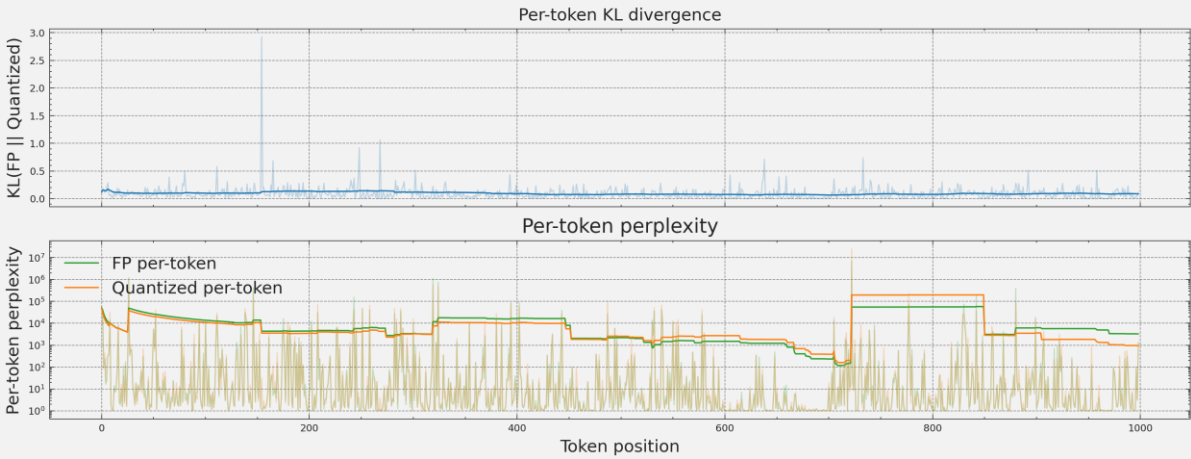
Relative error per token - Qwen3.5-0.8B @ 4 bits



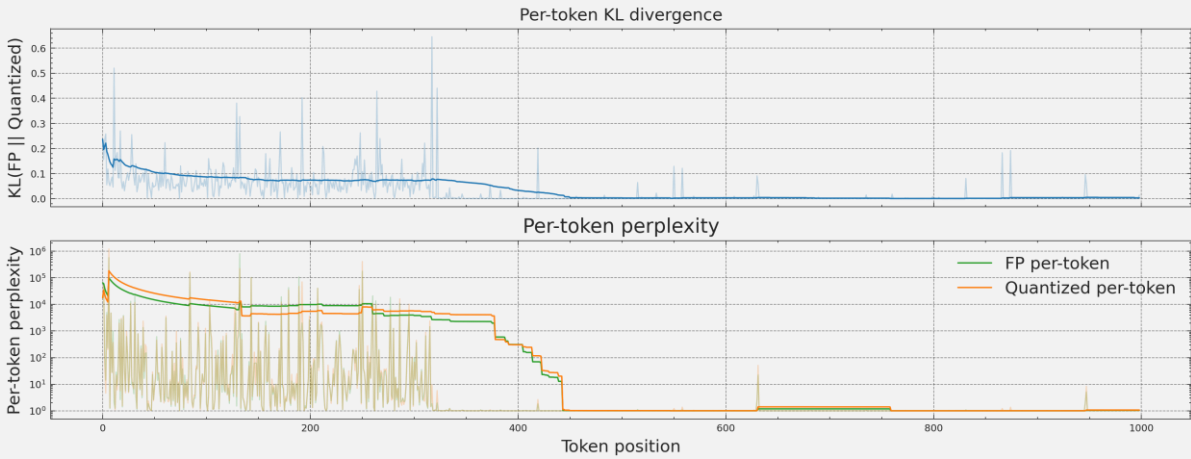
State drift

Mamba2-370m

Model: mamba2-370m-hf @ 4 bits on wikitext2_0



Model: mamba2-370m-hf @ 4 bits on custom_rep1



Next steps

1. Focus on Mamba-1.4b/-2.8b to understand their behaviour on repeated patterns (especially)
2. Continue on the **activations quantization**:
 - So far, tried quantizing only the inputs of the linear layers.
 - Quantize also the components of the SSM block (A,B,C,D matrices + time_step + hidden_states), and analyze their sensitivity
 - The state drift analysis can be re-applied also there